Exam 4 Mock Test

Name: _____ Section: ◯ 12 ◯ 29

Show all work to earn credit. 17 questions, 6 points each (102 total). Practice under exam conditions. Scientific calculators only; no graphing calculators, phones, or sharing calculators.

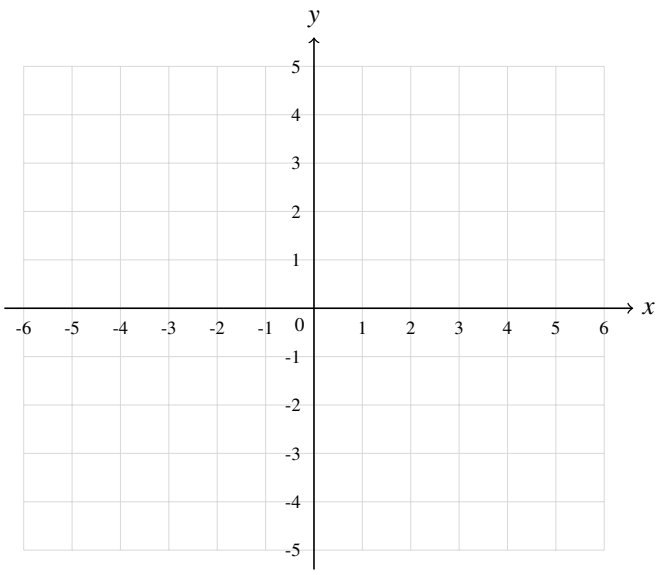
1. Find $\log_5\left(\frac{1}{125}\right)$.

2. Solve using the quadratic formula: $3x^2 - 2x - 2 = 0$.

3. Solve using the square root property: $(x + 2)^2 = 15$.

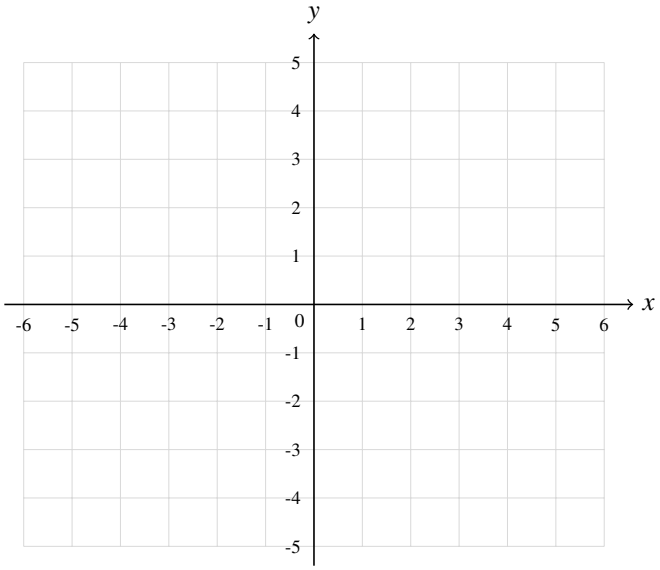
4. Solve using any method: $16x^2 + 16 = 0$.

5. Graph $f(x) = -x^2 + 4x + 1$. Give the vertex and axis of symmetry.



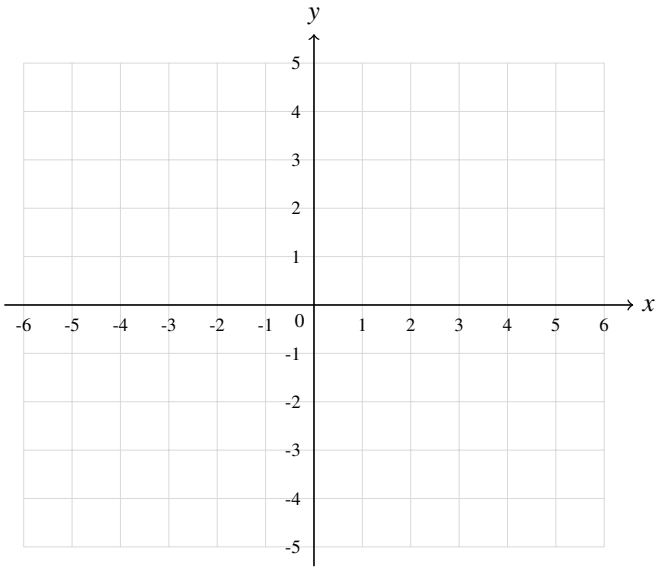
6. Let $f(x) = \sqrt{2x + 17}, \quad g(x) = x - 4$. Find and simplify $(f \circ g)(x)$.

7. Graph $f(x) = \log_3(x + 2)$. Plot two points and label the vertical asymptote.



8. Find the midpoint between $(-4, 6)$ and $(2, -2)$.

9. Graph $f(x) = \begin{cases} -2, & x < -1 \\ 2, & -1 \le x < 4 \\ 5, & x \ge 4 \end{cases}$. Indicate open and closed endpoints.

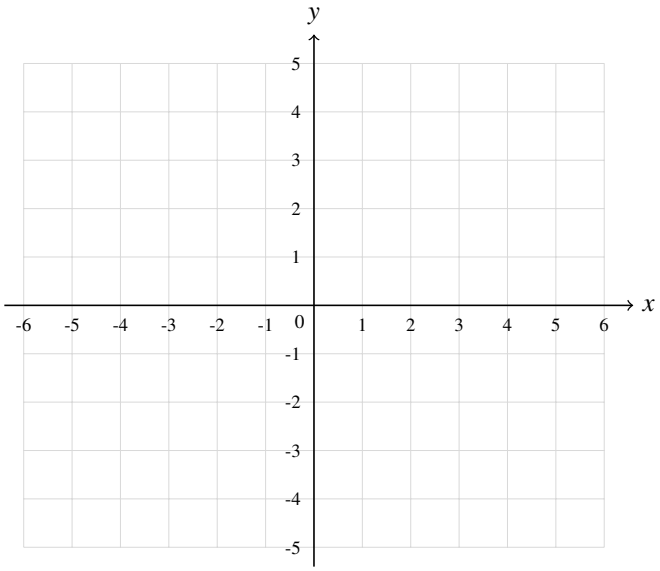


10. Find the exact distance between $(-1, -2)$ and $(5, 6)$. Simplify radicals.

11. A firecracker has height $h(t) = -5t^2 + 30t + 10$ feet after t seconds. Find its maximum height and the time it occurs.

12. Solve by completing the square: $x^2 + 4x - 2 = 0$.

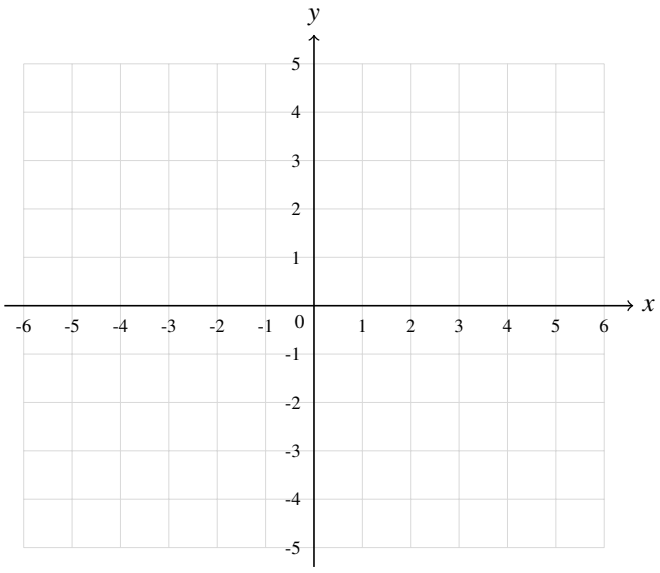
13. Graph $y = 2|x + 1| - 2$ using transformations.



14. Write $6^{-2} = \frac{1}{36}$ in logarithmic form.

15. Find the future value of \$4,000 at 5% for 6 years compounded continuously. Use $A = Pe^{rt}$. Round to cents.

16. Graph $f(x) = 3^x - 1$. Plot two points and label the horizontal asymptote.



17. Graph $f(x) = -(x + 1)^2 + 4$. Give the vertex and axis of symmetry.

